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PICKER ENGINEERING PROGRAM

Design Clinic
Project Summary Form

Sponsor: Honda Research Institute USA, Inc.
Location: 99P Labs, Columbus, Ohio
Timeframe: September 2026 – May 2027

Project Title: Design of a Sensor Suite for a Software-Defined Autonomous Vehicle Research Platform

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Project Scope/Overview:

The overall goal of this project is to transform a commercially available electric golf cart into a software-defined autonomous vehicle research platform capable of operating in three modes: manual human driving, remote human teleoperation, and bounded autonomous operation. Development of this vehicle will take please over multiple phases and subprojects. The aim of this capstone design project is focused on developing a comprehensive sensor suite to enable remote and autonomous operation. At the same time, a separate capstone team will be working on replacing the systems for driver-operated controls with systems capable of being electronically controlled (also known as x-by-wire). Pending the outcome of these two projects, future student teams will take the platform into the next phases.

The sensor suite may consist of various components, including Lidar, cameras, radar, and other environmental sensors, to provide the vehicle with the necessary situational awareness for safe autonomous operation. The first step in this project is to research, select, prototype, and calibrate the necessary sensors and as well as components needed for integration. The team will also develop initial framework for data collection, processing, and interpretation. Integration of these systems will allow for real-time data collection and analysis, critical for the vehicle's autonomous functions. The following outlines additional details for the anticipated design activities:

- **Sensor selection:** Identify and evaluate appropriate sensors, including Lidar, stereo cameras, GPS, radar, and internal measurement units (IMUs).
- **Sensor positioning:** Design and implement the physical mounting and connection of sensors to the body of the golf cart.
- **Testing and calibration:** Conduct extensive testing and calibration of all sensors to ensure functionality and reliability under various environmental conditions.
- **Data processing:** Develop a framework to format, standardize, and collect raw sensor data

This student team will be working simultaneously with a second capstone team on the autonomous golf cart. While each team will have their own distinct project objectives, there will be points at which collaboration is possible and necessary. In particular, it is anticipated that the two teams will collaborate on:

- **Centralize microcontrollers:** Implement a Single Board Computer (SBC) to centralize all processing and communication between the drive mechanisms and the sensor suite.
- **Integrate subsystem communication:** Implement a Controller Area Network (CAN) bus to allow for reliable communication between drive-by-wire and sensor subsystems.

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Agentic Workflow Augmentation Strategy:

A second objective of this project is to explore the use of multi-agent AI systems as engineering design collaborators, moving beyond simply using generative AI as a productivity aid. The team will develop a structured agentic workflow that incorporates various AI agents at different stages of the engineering process.

- **Research and analysis.** Research and critical-thinking agents may investigate sensor technology, safety features, and other relevant engineering literature. Specialized agents may independently analyze alternatives, identify conflicting evidence, document assumptions, and contribute to decision matrices.
- **Generative design.** AI CAD agents can be used for iteration within a constrained design space. For example, agents may generate or parameterize alternative actuator brackets, sensor mounts, electronics enclosures, or component layouts. Potential designs will be evaluated through conventional engineering analysis, simulation, inspection, and physical testing before acceptance.
- **Requirements management.** Agents may assist with requirements traceability, interface checking, test-case generation, and failure-mode brainstorming.

In all use-cases, students will remain responsible for verifying sources and approving conclusions. AI-generated recommendations will not constitute approval of safety-critical designs. Human review and conventional verification methods will remain authoritative.

As part of this investigation, the team will document where agentic methods provide measurable engineering value, where they fail or require substantial human intervention, and how agent-generated work can be incorporated responsibly into an engineering development process.

Disciplines involved: Electrical engineering, Mechanical engineering, Computer programming, Agentic AI

Project Background/Motivation:

The completed software-defined autonomous vehicle research platform will provide a common vehicle architecture for investigating how human and artificial-intelligence-based controllers perceive, decide, and act in controlled driving scenarios. Example measures may include reaction time, stopping distance, trajectory selection, obstacle clearance, and intervention frequency. The project is intended to establish the experimental platform and demonstrate representative comparisons rather than conduct a comprehensive human-subject research study.

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Sponsor Background:

Honda Research Institute, USA (HRI-USA) was established in 2003 as North America's advanced research center that provides innovative solutions to complex problems with direct applications to Honda's current and future technology roadmap. 99P Labs is part of HRI-USA with the mission to create sustainable mobility futures through data-driven innovation. 99P Labs helps innovators from business, government, and higher education come together to further the evolution and revolution of mobility and energy concepts. University students and Honda 99P interns advance innovations for future society at 99P through experiential learning and application of design thinking, data science, software engineering, and unique approaches to prototype testing.

Project Deliverables:

It is anticipated the student team will submit the following deliverables:

- Documentation of the project scope and design requirements for the complete sensor suite [Due late September as part of the Scope memo]
- A technical summary of sensors and detection technology relevant to autonomous vehicles [Due mid-October as the Technical Foundation memo]
- A list of individual sensors planned for the comprehensive suite, including multiple commercially available options for each sensor type [Due as part of Design Review 2]
- Evaluation of the proposed sensor suite including results from testing and calibration [Due mid-March as part of Design Review 4]
- Final prototype of the complete sensor suite, accompanied by engineering documentation test results, calibration data, electrical schematics, bill of materials, and any other relevant documentation [Due by the end of Q4]
- Reflection on the use of agentic AI workflows within the design process [Due by the end of Q4 as part of the final report]